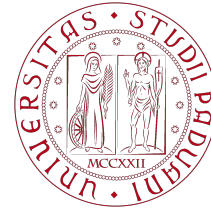


Final Report

Physics of Complex Networks: Structure and Dynamics



UNIVERSITÀ
DEGLI STUDI
DI PADOVA

Areas of physics by complexity



Newton's
Mechanics

Electro-
Magnetism

Special
Relativity

Quantum Mechanics
General Relativity

Quantum
Field Theory

Complexity
Science

Projects Report: #15, #47

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Last update: August 22, 2026

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1 | Task #15: Cascading Failures - SOC model

1.1 | Introduction

Self-Organized Criticality (SOC) is an approach adopted to explain the emergence of power-law distributions in natural phenomena. In physics, criticality is associated to systems near phase transitions, that live between two distinct regimes, and at this critical point, the emergence of power-laws occurs.

In SOC, systems are believed to organize themselves naturally into critical states, following internal feedback mechanisms and without external intervention.

In this report, we analyze the SOC behavior associated to the Sandpile Model.

In the **Sandpile Model**, we consider networks with an associated load variable, that accounts for any sort of stress that can be observed in real applications. A given node in the network stays functional, as long as its load doesn't go above a given threshold: its *capacity*. Load is assigned randomly to nodes in the network, and when the capacity of a node is overcome, we have a failure in the node, and its load is redistributed according to some chosen rules. A dissipative factor is added to load transfers, to avoid excessive accumulation of load across the network.

We use single-layer¹ and two-layer² synthetic networks, where the first ones are used to examine the basic properties of cascade failures, and the second offers a more robust study of interconnected systems, such as real power grids.

To analyze the cascades, we assume a tree structure of each avalanche, and using the theory of multiplicative branching processes, we find analytical predictions for the distributions of the size of the avalanches $p(s)$ and of their lifetime $l(t)$.

Considering:

$$p(s) \sim s^{-\tau} \tag{1.1}$$

$$l(t) \sim t^{-z} \tag{1.2}$$

The results for the single layer systems studied in this work are reported in Tab. 1.1. For interconnected networks, the analytical results for each single layer are not obtained,

¹Following the works [1] and [3].

²Following [2].

but considering the distributions of cascades over the whole network, the results are equivalent to a random network.

Network	τ	z
Random	$3/2$	2
$2 < \gamma < 3$	$\frac{\gamma}{\gamma-1}$	$\frac{\gamma-1}{\gamma-2}$
Scale-free (γ) $\gamma = 3$	$3/2^*$	2^*
$\gamma \geq 3$	$3/2$	2

Table 1.1: Analytical predictions for avalanche distributions.

* with logarithmic corrections.

1.2 | Sandpile Model on Single Layer Networks

We first consider the Sandpile Model on single layer networks with simple degree distributions, in particular, following the works of Bonabeau [1] and Goh *et al.* [3], we use the distributions:

- Gaussian (Random): $k \sim \mathcal{N}(\mu = 20, \sigma = 8)$
- Uniform (Random): $k \sim \text{Uniform}(k_{\min} = 4, k_{\max} = 36)$
- Power Law (Scale Free): $k \sim k^{-\gamma}$ with $\gamma \in \{2.01, 2.2, 2.4, 2.6, 2.8, 3.0, 5.0\}$

Every network includes $N = 10^5$ nodes, and the results are the averages of 30 different iterations. The dissipation is carried by dissipative nodes (0.1% of the total) in the random networks, which transfer load outside of the network once toppled; we use a dissipation probability $f = 0.001$ on the scale-free network, which is the probability of every load transfer to not happen, dissipating the load.

Fig. 1.1 displays the results obtained for avalanche size and lifetime distributions in the random networks. Excluding dissipation effects, visible on the heads of the distributions, and finite-size effects, visible on their tails, the results follow the analytical prediction, shown in the dashed line.

The results for the scale-free networks are reported in Section 1.4, along with a brief analysis on the exponents extracted from the distributions.

1.3 | Interconnected Networks

We conclude the analysis by studying how interconnectivity between two networks affects cascading failures, following the work of Brummitt *et al.* [2]. Real power grids are sparsely interconnected with narrow degree distributions; to model this we consider two independent random 3-regular graphs, each with $N = 2 \times 10^3$ nodes, coupled by a *Bernoulli interconnection*: each node of either network independently receives one additional external edge stub with probability p (and none with probability $1 - p$). The external stubs of the two networks are then matched uniformly at random, so that both networks have the same number of interconnections. We denote this topology by $R(3)-B(p)-R(3)$; the threshold of each node is its total degree, as before.

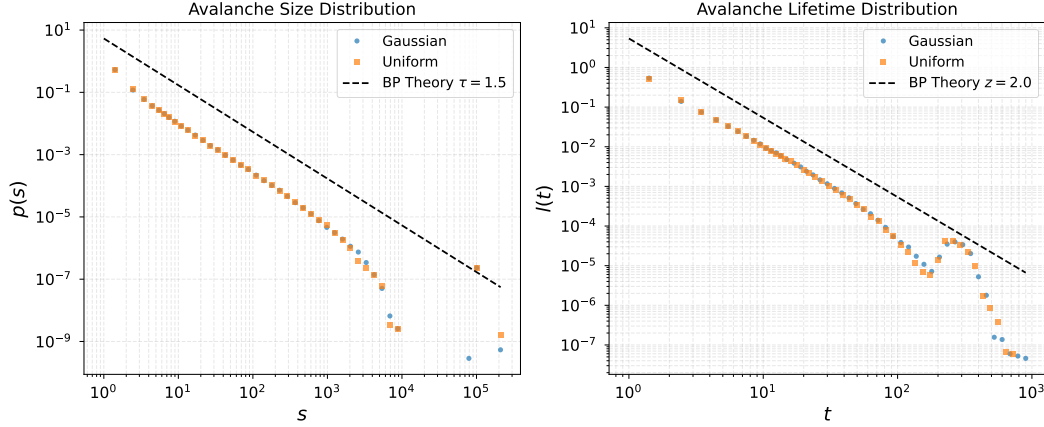


Figure 1.1: Avalanche size (s) and lifetime (t) distributions for the sandpile model on Gaussian and Uniform networks, along with the expected theoretical results (dashed lines).

We run the sandpile dynamics with dissipation rate $f = 0.01$, chosen so that the largest cascades topple almost the entire network, dropping 2×10^6 grains per realization (after a 20% transient) on 30 independent realizations for 10 values of p in $[0.001, 0.5]$, concentrating on small p as in the reference work.

We define T_o as the number of topplings caused in network $o \in \{a, b\}$, and $T_{m,n}$ as the number of topplings in network n caused by a cascade started in network m .

Figure 1.2 shows the central result: adding a few interconnections between the networks is *locally stabilizing*, the probability of a large avalanche in a ($T_a > N/2$) initially drops with p , because excess load is absorbed by the neighboring network, which acts as a reservoir. However, beyond a critical amount of interconnectivity the probability rises again, for two reasons: interconnections open pathways for network b to inflict large cascades onto a (the red curve only ever increases), and every added interconnection augments the total capacity and average load of the system, fueling larger cascades. The two effects balance at $p^* \approx 0.075$, where the probability of a large avalanche in a is minimized.

The effects of increased connectivity on global stability and different avalanche sizes are investigated in Section 1.4.

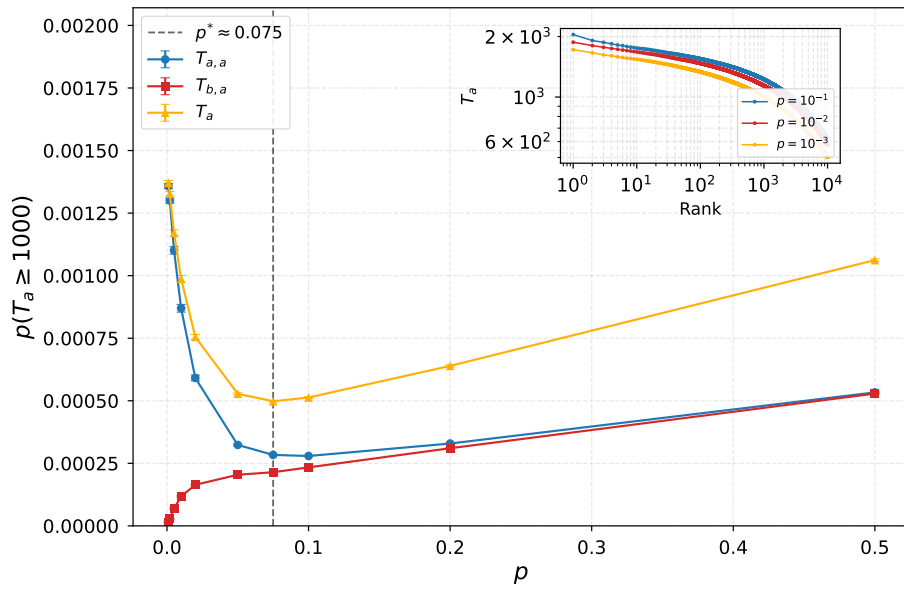


Figure 1.2: Probability of a large avalanche ($T_a > 1000 = N/2$) in network a as a function of the interconnectivity p . Large cascades *beginning* in a (blue) first decrease and then increase with p , whereas cascades beginning in b and toppling most of a (red) become monotonically more likely. Large cascades in a are minimized at $p^* \approx 0.075$. Inset: log-log rank-size plots of the 10^4 largest values of T_a for $p = 10^{-1}, 10^{-2}, 10^{-3}$, showing the suppression of the largest local cascades as interconnectivity is initially increased.

1.4 | Supplementary Material

1.4.1 Scale-Free Networks

Fig. 1.3 displays the avalanche size and lifetime distributions for the scale-free networks, at the chosen gamma values. The results follow the theoretical predictions for $\gamma \geq 3$.

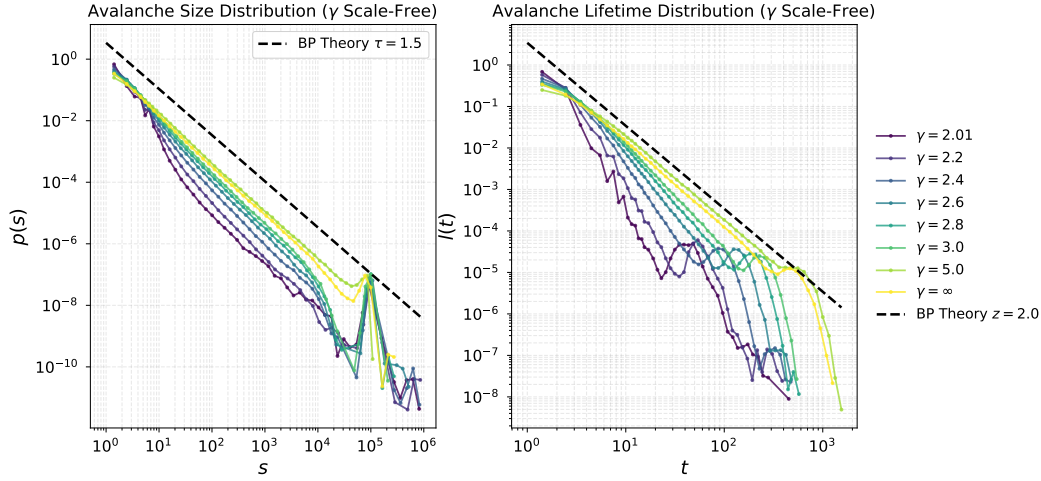


Figure 1.3: Avalanche size (s) and lifetime (t) distributions for the sandpile model on Scale-Free networks, along with the expected theoretical results for $\gamma \geq 3$ (dashed lines).

1.4.2 Exponents Analysis

To analyze other gamma values, we estimate the scaling exponents fitting the observed distributions, and report the results in Tab. 1.2.

For each distribution the fitting window is chosen as the longest straight window: sliding a lower cutoff over a set of candidates and the upper edge is extended as far as the weighted least-squares fit maintains a coefficient of determination³ $R^2 \geq 0.999$ for the avalanche sizes and $R^2 \geq 0.998$ for the (noisier) lifetimes. Finally, the window with the largest logarithmic span is selected. The window chosen for each fit is reported in the Table.

The error bars in the exponents are the standard deviation of the exponent over the family of plausible windows obtained by varying the R^2 threshold and the lower cutoff by one step. This systematic uncertainty, reflecting how much the exponent changes under a relocation of the fitting window, dominates the scatter of the regressions.

The fitted exponents are in good agreement with the multiplicative branching process predictions: on the random networks the lifetime exponent matches $z = 2$ within the window-sensitivity error, and the size exponent sits only slightly above the mean-field value $\tau = 3/2$ (a finite-size effect visible in the mild slope drift of the local slopes). For the scale-free networks the observed exponents decrease with γ as predicted, although

³ $R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$, where $SS_{res} = \sum_i (y_i - f(x_i))^2$ is the squared residuals between data y_i and predictions $f(x_i)$; and $SS_{tot} = \sum_i (y_i - \bar{y})^2$ is a coefficient proportional to the variance of the dataset.

Network	τ		z		s		t	
	theory	fit	theory	fit	min	max	min	max
Gaussian	1.50	1.62 ± 0.01	2.00	1.98 ± 0.06	5	3e+03	4	6e+01
Uniform	1.50	1.61 ± 0.01	2.00	2.00 ± 0.05	5	3e+03	4	6e+01
$\gamma = 2.01$	1.99	1.75 ± 0.10	101.00	5.58 ± 0.21	58	4e+02	54	1e+02
$\gamma = 2.2$	1.83	1.81 ± 0.06	6.00	5.90 ± 0.33	58	3e+03	68	2e+02
$\gamma = 2.4$	1.71	1.76 ± 0.03	3.50	3.44 ± 0.05	58	1e+04	6	6e+01
$\gamma = 2.6$	1.62	1.79 ± 0.04	2.67	2.90 ± 0.03	20	1e+04	4	1e+02
$\gamma = 2.8$	1.56	1.75 ± 0.02	2.25	2.62 ± 0.04	20	6e+04	4	1e+02
$\gamma = 3.0$	1.50	1.72 ± 0.03	2.00	2.42 ± 0.03	17	2e+04	4	2e+02
$\gamma = 5.0$	1.50	1.60 ± 0.00	2.00	1.95 ± 0.09	5	3e+04	4	7e+02
$\gamma = \infty$	1.50	1.63 ± 0.01	2.00	2.03 ± 0.02	5	6e+04	4	5e+02

Table 1.2: Fitted avalanche-size (τ) and lifetime (z) exponents vs multiplicative branching-process predictions, together with the scaling ranges used for each fit. Fits are performed on the longest straight window of each log-binned distribution and error bars are the window-sensitivity of the fit (see text). The corresponding fit windows are listed, s refers to size intervals and t to lifetime intervals.

the fitted τ remains below the theoretical value for $\gamma = 2.01$ and rises above it for $\gamma \gtrsim 2.4$; for γ close to 2 the theoretical lifetime exponent diverges as $z = (\gamma - 1)/(\gamma - 2)$, which is not attainable in finite networks, so the comparison there is only indicative.

1.4.3 Interconnected Networks

The locally stabilizing effect is specific to each individual network; the collection of networks viewed as one system is instead *globally destabilized* by interconnectivity. Figure 1.4 shows that the total avalanche size distribution $p(s)$ (considering topplings in both networks) develops a longer tail as p grows, and the inset confirms that the largest global cascades become larger with p : while a single network benefits from interconnectivity, the whole system suffers larger cascades.

Figure 1.5 shows that the effect of interconnectivity depends on the size of the cascade considered: small cascades ($1 \leq T_a \leq 51$) are favored by larger p , intermediate cascades ($100 \leq T_a \leq 150$) occur most frequently at an unstable critical point around $p^* \approx 0.075$, while the largest cascades ($650 \leq T_a \leq 700$ and $850 \leq T_a \leq 900$) are suppressed at the same equilibrium point p^* . A network cannot simultaneously mitigate all cascade sizes: interconnectivity that minimizes the largest cascades amplifies the intermediate ones, and vice versa.

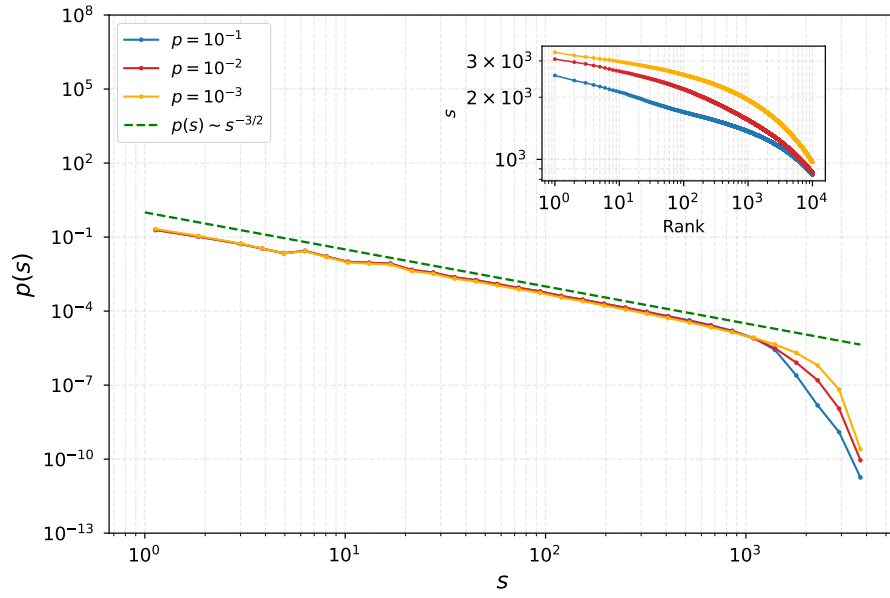


Figure 1.4: Total avalanche size distribution $p(s)$ (topplings in both networks) on $R(3)-B(p)-R(3)$ for $p = 10^{-3}, 10^{-2}, 10^{-1}$, in log-log. The green dashed line is the mean-field prediction $p(s) \sim s^{-3/2}$ obtained when both networks are viewed as a single system. Inset: log-log rank-size plot of the 10^4 largest global cascades, showing that increasing p enlarges the largest cascades of the whole system due to the additional interconnections.

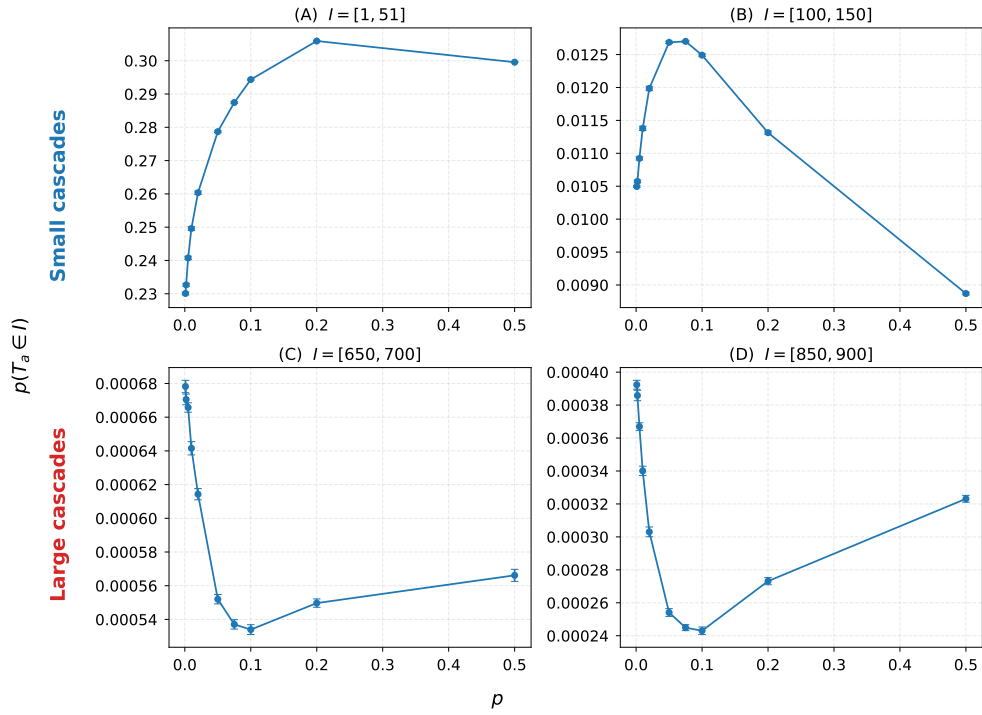


Figure 1.5: Probability that a cascade in network a has total size T_a within each of four windows, as a function of p on $R(3)$ - $B(p)$ - $R(3)$. Small cascades (A) become more likely with p ; intermediate cascades (B) peak at an unstable critical point near $p^* \approx 0.075$; the largest cascades (C, D) are minimized at the stable equilibrium $p^* \approx 0.075$. Tuning the interconnectivity to suppress one range of cascade sizes amplifies the others.

2 | Task #47: Spatial Covariance and Landau-Ginzburg Models for Embedded Empirical Networks

2.1 | Introduction

Statistical field theory offers a principled way to describe long-range correlations in spatially extended systems through a small set of potential coefficients. Extending these ideas to networks is difficult, but *spatially embedded* networks provide a natural starting point: their topology can be summarized by a *graph kernel*: the probability $p(r)$ that two nodes separated by a distance r are connected, which plays the role of a two-point correlation function, and can be compared with the covariance structure of a statistical field defined over the same space.

We consider a scalar Landau-Ginzburg (LG) theory with free energy

$$\mathcal{F}[\phi] = \int dx \left[\frac{\kappa}{2} (\nabla \phi)^2 + \frac{r_0}{2} \phi^2 + \frac{u}{4} \phi^4 \right], \quad (2.1)$$

with statistical weight $P[\phi] \propto e^{-\mathcal{F}[\phi]/T}$. In the Gaussian case ($u = 0$) the connected covariance is $G = TQ^{-1}$, where $Q = -\kappa \nabla^2 + r_0$ is the precision operator and $\xi = \sqrt{\kappa/r_0}$ the correlation length. We discretize the spatial domain on a regular grid of spacing h and build Q as the symmetric matrix

$$Q = \frac{\kappa}{h^2} L + r_0 I, \quad (2.2)$$

where L is the five-point graph Laplacian and I the identity; the covariance is then obtained by sparse matrix inversion, $G = TQ^{-1}$, and its radial profile is compared with the empirical kernel. The task is therefore to test whether the empirical graph kernel can be reproduced by such a field theory, at three levels of description: the empirical kernel fitted directly from the data, a *homogeneous* LG model (global coefficients), and a *non-homogeneous* LG model whose mass varies in space, $r(x) = r_0(1 + \varepsilon m(x))$, with $m(x)$ the local node-density anomaly.

2.2 | Data and Methods

We use the GridKit dataset¹ of European and North-American high-voltage power-grid extracts, with stations as vertices and power lines as links. Node coordinates are projected to an equirectangular metric system (centered on the network centroid, $R = 6371$ km). After collapsing multi-edges we restrict each network to its largest connected component, obtaining $N = 13\,478$ nodes and $E = 16\,922$ links for Europe, and $N = 14\,990$ nodes and $E = 18\,804$ links for North America ($\langle k \rangle \approx 2.5$).

The homogeneous graph kernel is estimated by binning node pairs into logarithmically spaced radial bins and taking $p(r) = \text{links/pairs}$ in each bin. We fit a family of parametric kernels (exponential, Gaussian, power-law with cutoff, stretched exponential, and their product) by binomial log-likelihood over the bins, selecting models by AIC. For the LG side we discretize the spatial domain, build the quadratic precision operator and obtain the covariance G by sparse matrix inversion; its radial profile is averaged over source columns drawn from the grid interior and binned by distance with multiplicity weighting, and fitted to the empirical kernel by weighted least squares in log-log space. We fit the amplitude A and the mass r_0 ; the stiffness and temperature are set to $\kappa = T = 1$: they do not change the radial functional form, since they are absorbed by other parameters; so the single shape parameter is the correlation length $\xi = \sqrt{\kappa/r_0} = r_0^{-1/2}$. For the non-homogeneous level the mass varies over coarse spatial regions, $r(x) = r_0(1 + \varepsilon m(x))$ with $m(x)$ the node-density anomaly: we split the network into dense and sparse regions, fit the operator to each region's radial kernel separately (obtaining $r_{\text{dense}}, r_{\text{sparse}}$), and invert the two relations $r_g = r_0(1 + \varepsilon m_g)$ for the coefficients (r_0, ε) . The quartic coupling is kept at $u = 0$ (Gaussian baseline); a nonzero u would merely renormalize the mass, an approximation we document in Sec. 2.5, together with a more in-depth non-homogeneous analysis.

2.3 | Results

Figure 2.1 shows the homogeneous graph kernel together with the parametric fits and the LG reconstruction for the European network (North-American results reported in Fig. 2.2). The best parametric model for both networks is an exponential times power law, $p(r) \propto e^{-r/\ell}(1 + r/r_0)^{-\alpha}$, (fit results in Table 2.1), preferred over the exponential and the pure power-law cutoff by AIC (Table 2.2).

The homogeneous LG model reproduces the kernel well at high and intermediate distances ($r \geq \xi$), but it is not representative at lower distances (see Figs. 2.1 and 2.2). This model is limited by the grid spacing $h = 15$ km, and it can predict only above this distance; furthermore, the fitting of the model is based on Least Squares, weighted on pair counts. Since the number of pairs $\propto r^2$, the tail dominates the fit, and the smaller data are less represented. The discrete covariance of the massive Gaussian operator decays as $e^{-r/\xi}/\sqrt{r}$, falling faster than the empirical power-law tail. The shape mismatch would be present also changing the grid spacing to smaller values (at the price of a massive increase in computational cost of the fit), and this is the signature that a Gaussian field theory with a single mass cannot capture the heavy-tailed spatial organization of the power grid.

¹B. Wiegman, *GridKit: European and North-American extracts*, Zenodo, 10.5281/zenodo.47317 [4].

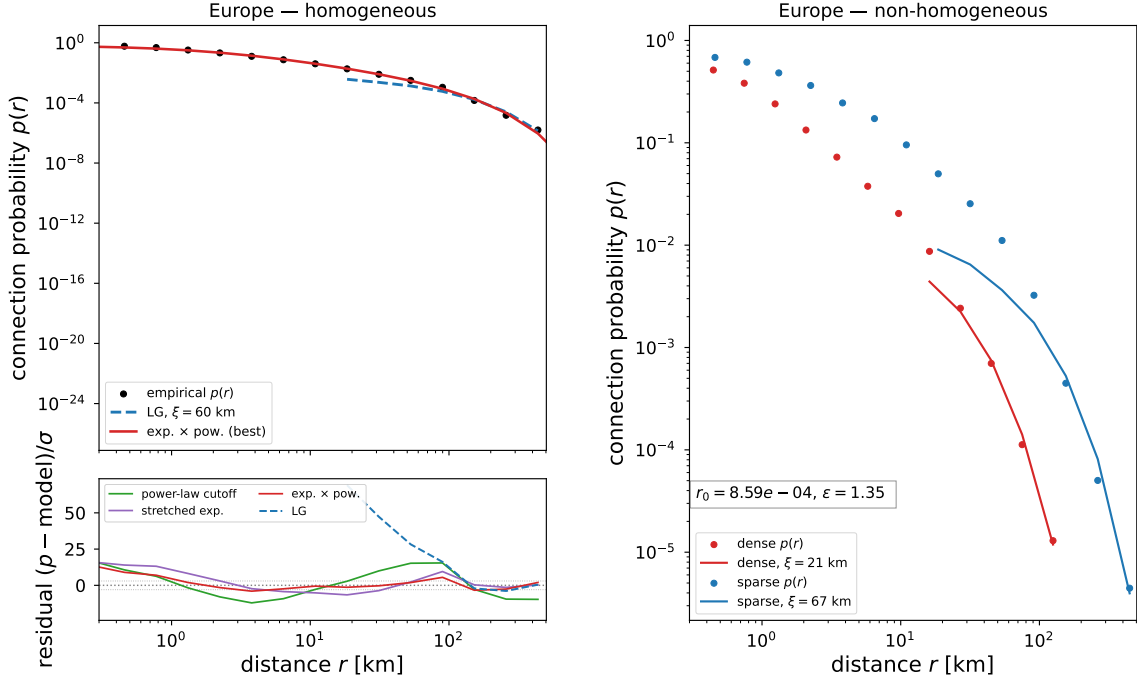


Figure 2.1: Fit results for the European network. Left: empirical homogeneous kernel $p(r)$ with the best parametric fit and the homogeneous Landau-Ginzburg operator $Q = (\kappa/h^2)L + r_0I$ (dashed); below it, the standardized residual $(p - \text{model})/\sigma$ (per-bin binomial standard error σ) for the power-law-capable kernels (power-law cutoff, stretched exponential, exponential-times-power-law) and the operator, with a zero line and a $\pm 3\sigma$ band; the pure exponential-decay kernels (exponential, Gaussian) are off-scale in the tail and omitted. Right: the non-homogeneous decomposition into dense (red) and sparse (blue) regions, each with its own operator fit; the inferred non-homogeneous coefficients (r_0, ε) are reported. North American results shown in Fig. 2.2.

The non-homogeneous level quantifies the spatial heterogeneity directly: dense regions are characterized by a much shorter correlation length than sparse ones, $\xi_{\text{dense}} \approx 21$ km versus $\xi_{\text{sparse}} \approx 67$ km for Europe. Inverting the mass relation gives $r_0 \approx 8.6 \times 10^{-4}$ and $\varepsilon \approx 1.35$ (Europe); a strongly density-dependent mass with $\varepsilon > 0$: urbanized regions (positive density anomaly) carry a larger mass and hence connect over shorter scales, while rural regions sustain long transmission lines. The North-American decomposition shows the same qualitative pattern and is reported in the Supplementary Material. The density maps (Fig. 2.3) confirm that the dense/sparse split corresponds to a genuine geographic, not merely statistical, partition.

2.4 | Discussion

The three levels of description form a coherent hierarchy. The empirical graph kernel establishes a heavy-tailed, power-law baseline (an exponential times power-law decay is preferred by AIC). The homogeneous LG model captures the correct correlation length but, being a massive Gaussian theory, is constrained to exponential decay and therefore underestimates the tail. The non-homogeneous model, with a density-dependent mass, recovers the local correlation structure and identifies dense and sparse regions as carrying distinct effective masses. In this sense the field-theoretic description does capture nontrivial spatial organization beyond a simple distance-decay kernel: the dependence of the correlation length on local density is precisely the kind of information a single homogeneous kernel discards.

Several approximations limit the analysis and are documented explicitly. (i) The equirectangular projection is adequate at continental scale but not conformal; a proper projected CRS would refine the distance estimates. (ii) The discrete operator has no covariance for distances below the lattice spacing h : the ultraviolet divergence of the continuum Green's function is regularized by h , so the model is only matched for r above a cutoff $r_{\min} = h$. The empirical kernel is nevertheless measured below h (down to ~ 0.1 km), so the smallest bins are shown but not described by the field theory. (iii) The quartic term is set to $u = 0$; treating $u > 0$ (mean-field or perturbatively) would renormalize r_0 but not alter the qualitative exponential decay. (iv) Power grids are quasi-one-dimensional (they follow geographic corridors), so the isotropic two-dimensional covariance is an idealized reference rather than an exact model. Finally, the non-homogeneous inference is performed with a two-region density split; a continuous $m(x)$ over a finer spatial grid is a natural extension.

2.5 | Supplementary Material

2.5.1 Fit Results

Network	ℓ [km]	r_0 [km]	α
Europe	75	1.9	1.4
North America	81	2.7	1.6

Table 2.1: Fit parameters for the exponential times power law model (best model) for both networks.

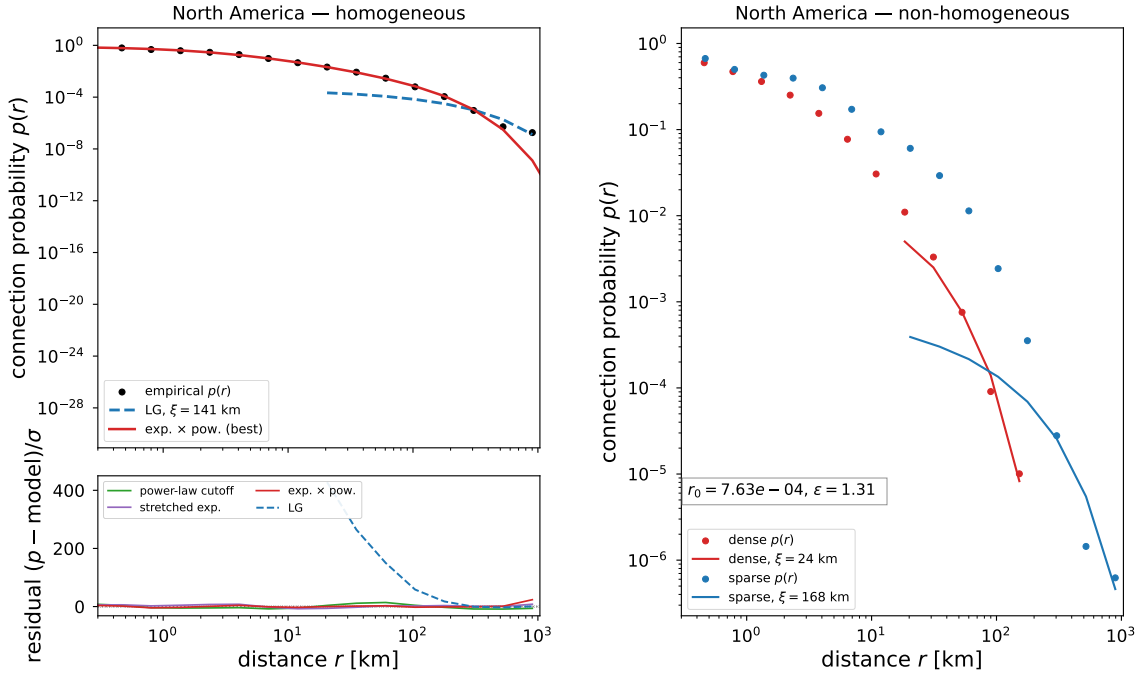


Figure 2.2: Fit results for the North American network. Left: empirical homogeneous kernel $p(r)$ with the best parametric fit and the homogeneous Landau-Ginzburg operator $Q = (\kappa/h^2)L + r_0I$ (dashed); below it, the standardized residual $(p - \text{model})/\sigma$ (per-bin binomial standard error σ) for the power-law-capable kernels (power-law cutoff, stretched exponential, exponential-times-power-law) and the operator, with a zero line and a $\pm 3\sigma$ band; the pure exponential-decay kernels (exponential, Gaussian) are off-scale in the tail and omitted. Right: the non-homogeneous decomposition into dense (red) and sparse (blue) regions, each with its own operator fit; the inferred non-homogeneous coefficients (r_0, ϵ) are reported.

The homogeneous LG operator gives $\xi \approx 141$ km with $R^2 = 0.59$, a weaker fit than Europe because the network spans larger distances and its power-law tail is more prominent (Table 2.3). The non-homogeneous split yields $\xi_{\text{dense}} \approx 24$ km versus $\xi_{\text{sparse}} \approx 168$ km, inverting to $\epsilon \approx 1.31$.

²Given a statistical model with k estimated parameters, the Akaike Information Criterion is defined as $\text{AIC} = 2k - 2\ln(L)$, where L is the maximized value of the Likelihood function for the model.

Model	DOF	NLL (EU)	AIC ² (EU)	NLL (NA)	AIC (NA)
exponential	2	6.681×10^4	1.336×10^5	6.959×10^4	1.392×10^5
gaussian	2	7.649×10^4	1.530×10^5	8.029×10^4	1.606×10^5
power-law cutoff	3	6.033×10^4	1.207×10^5	6.300×10^4	1.260×10^5
stretched exp.	3	6.006×10^4	1.201×10^5	6.274×10^4	1.255×10^5
exp. $\times$ pow.	4	5.974×10^4	1.195×10^5	6.258×10^4	1.252×10^5

Table 2.2: Model selection for the homogeneous graph kernel (binomial log-likelihood). The exponential-times-power-law model is preferred for both networks. Negative Log Likelihood (NLL) and AIC are reported as fit quality estimates.

Network	ξ (km)	r_0^{hom} (km ⁻²)	$r_0^{\text{non-hom}}$ (km ⁻²)	ξ_d (km)	ξ_s (km)	ε
Europe	60.1	2.77×10^{-4}	8.59×10^{-4}	21.4	66.5	1.35
North America	140.7	5.05×10^{-5}	7.63×10^{-4}	24.0	167.9	1.31

Table 2.3: Inferred Landau-Ginzburg parameters. Homogeneous: correlation length ξ and mass $r_0^{\text{hom}} = \kappa/\xi^2$ ($\kappa = 1$), fitted from the discretized operator $Q = (\kappa/h^2)L + r_0I$. Non-homogeneous: density-conditioned correlation lengths and the baseline mass of $r(x) = r_0^{\text{non-hom}}(1 + \varepsilon m(x))$, together with the coupling ε .

2.5.2 Node Density of the networks

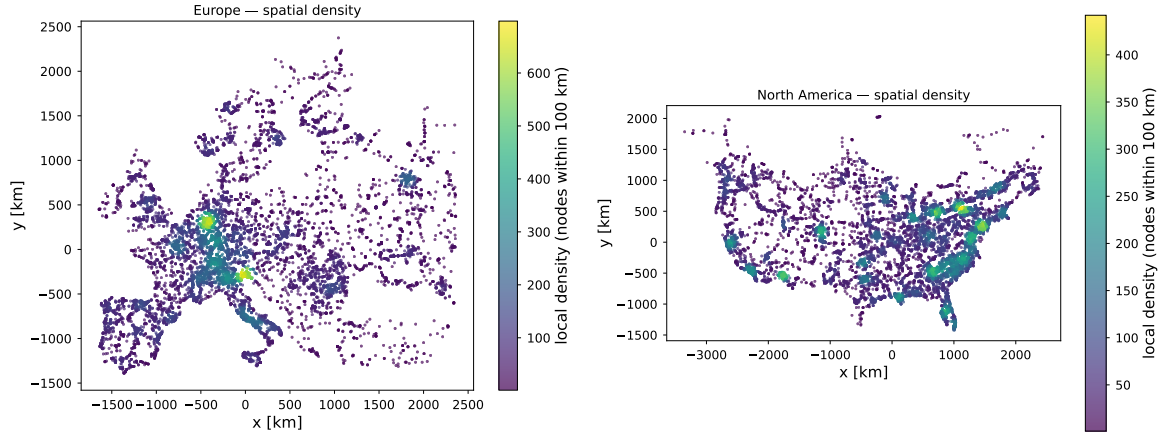


Figure 2.3: Local node density (neighbours within 100 km) for Europe (left) and North America (right). The dense/sparse partition used for the non-homogeneous LG model is a genuine spatial structure.

2.5.3 Perturbative extension: non-homogeneous mass and quartic coupling

This section tries to make perturbative level precise. *What is computed:* (i) the first-order (in ε) correction δG to the covariance induced by a spatially varying mass, and (ii) the one-loop (in u) mass renormalization of the quartic vertex. *What analysis is performed:* the ε -expansion is validated against the exact inversion (truncation error

$\propto \varepsilon^2$), the radial average of δG is shown to be suppressed (so ε is identified from the density-conditioned kernels rather than the radial profile), and the vacuum variance $\langle \phi^2 \rangle_0$ is evaluated to bound the admissible u . *What results are reached:* a positive coupling $\varepsilon \approx 0.83$ (Europe) and 0.44 (North America), consistent in sign with the two-region inversion ($\varepsilon \approx 1.3$); a UV-sensitive $\langle \phi^2 \rangle_0 \approx 111/142$; and a quartic coupling $u \sim 8 \times 10^{-7}$ that already shifts the mass by r_0 , showing the ϕ^4 vertex is strongly relevant in $d = 2$ while leaving the exponential tail qualitatively unchanged.

The non-homogeneous mass $r(x) = r_0(1 + \varepsilon m(x))$ and the quartic coupling u are treated perturbatively around the homogeneous Gaussian operator, so that the precision matrix is factorized *once* and every parameter value is explored without a further decomposition. The local density anomaly is discretized on the same regular grid: the number of *network vertices* within 100 km of each grid site (a KD-tree range query, the same estimator used for the node-level density, standardized over grid sites), giving a continuous field $m(x)$. The precision matrix splits as

$$Q(\varepsilon) = Q_0 + \varepsilon M, \quad M = r_0 \text{diag } m(x), \quad Q_0 = \frac{\kappa}{h^2} L + r_0 I, \quad (2.3)$$

and the covariance expands as $G = TQ^{-1} = G_0 + \delta G + \mathcal{O}(\varepsilon^2)$, where

$$\delta G = -\varepsilon T Q_0^{-1} M Q_0^{-1}, \quad \text{or, per source column,} \quad \delta G e_s = -\varepsilon Q_0^{-1} (r_0 m(x) \odot g_0), \quad (2.4)$$

with $g_0 = TQ_0^{-1}e_s$ (the sampling temperature T cancels). A single sparse LU factorization of Q_0 therefore supplies the $\mathcal{O}(\varepsilon)$ correction at the cost of one extra back-substitution per source column. The radial average of δG is strongly suppressed: the anomaly has zero mean, so only the spatial variation of the smoothing kernel contributes (the radial correction reaches $\approx 12\%$ of the peak for Europe at the fitted ε ; Fig. 2.1); this is precisely why ε is identified from a *density-conditioned* observable (the dense- and sparse-restricted kernels) rather than from the radial kernel alone. The truncation is second order: at $\varepsilon = 0.05$ the perturbative profile differs from the exact inversion by $\lesssim 2 \times 10^{-5}$ relative to the peak, growing $\propto \varepsilon^2$.

Fitting (A, ε) (with r_0 fixed at its homogeneous value) to the dense- and sparse-conditioned kernels gives $\varepsilon \approx 0.83$ (Europe) and $\varepsilon \approx 0.44$ (North America), consistent in sign with the two-region inversion ($\varepsilon \approx 1.3$): a positive density anomaly carries a larger effective mass. The smaller magnitude reflects two documented approximations: the fit uses a single global amplitude A , whereas dense and sparse populations have markedly different overall connection rates, and the standardized grid field $m(x)$ additionally samples empty space that the node-level anomaly does not.

To first order in u the only effect of the ϕ^4 term is a renormalization of the mass,

$$r_{\text{eff}} = r_0 + 3u \langle \phi^2 \rangle_0, \quad \langle \phi^2 \rangle_0 = T \frac{1}{N} \text{tr } Q_0^{-1}, \quad (2.5)$$

the factor 3 counting the tadpole contractions of the $u/4 \phi^4$ vertex. The vacuum variance $\langle \phi^2 \rangle_0$ is ultraviolet-sensitive: on the lattice it evaluates to ≈ 111 (Europe) and ≈ 142 (North America). The quartic coupling is therefore strongly relevant in $d = 2$: a shift as large as $\delta r = r_0$ already requires $u \sim r_0 / (3 \langle \phi^2 \rangle_0) \approx 8 \times 10^{-7}$. This makes quantitative the statement of the main text that $u > 0$ only renormalizes the mass (shrinking $\xi = \sqrt{\kappa/r_{\text{eff}}}$) without altering the qualitative exponential decay.

3 | Use of AI tools

This chapter documents how generative AI tools (large language models) were used during the development of Task #15 and Task #47, in compliance with the course policy on AI-assisted work. The tools were used as a coding and writing assistant; every result was reviewed and validated by the author.

In Task #15, AI was used mainly as a coding tool, where every part of the code was specifically tuned for the needs I (human) required.

On the other hand, in Task #47 I wanted to explore more deeply the world of agentic AI: I used an Opencode environment specifically tuned to work with the project constraints given by the professor. This gave me full control of the *ideas* behind the code, while losing direct control of the code itself. At the same time, I could invest more energy in the concepts and difficult mathematical theories behind this project.

In both tasks, AI was used as a writing assistant, to help correct syntax and typos.

4 | Bibliography

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